Phronesis

Adaptive AI practice to rebuild code comprehension

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Area of Investigation: AI is outpacing understanding

Vibe coding: using artificial intelligence, prompted by natural language, to assist with writing computer code (Collins Dictionary, 2025)

84%
of developers use or plan to use AI tools, 51% of professionals daily

(Stack Overflow, 2025)

50% vs 67%
comprehension quiz score, AI-assisted vs hand-coding developers, biggest gap on debugging

(Shen and Tamkin, 2026)

17% lower
grades once AI access was removed, for students with unrestricted GPT-4

(Bastani *et al.*, 2025)

How AI is used matters: designs that keep learners engaged largely avoided the harm (Bastani *et al.*, 2025; Shen and Tamkin, 2026)

Gap in current solutions

What exists	What it offers	What it misses
Interview platforms (e.g. LeetCode)	Huge fixed problem bank	Built around passing interviews
AI add-ons (e.g. LCPath, LeetCode Coach)	Weakness analysis and hints (LeetCode Coach, 2026)	Aimed at interview candidates, still centred on solving problems
Course platforms (e.g. Codecademy)	AI tutoring	Set curriculum
Knowledge tracing research	Models learner progress over time	Mostly predicts correctness from quiz-style data (Dai *et al.*, 2021)

The gap: adaptive practice aimed at comprehension (reading, tracing and debugging code). Tracing and explaining skills are linked to code writing ability (Lopez *et al.*, 2008)

Proposed AI solution

- short daily challenge set tailored to your weak concepts
- weakness model: mastery score per concept (e.g. knowledge tracing)
- llm: generates variants and hints
- architecture diagram goes here

Prototype and feasibility

```
def next_challenge(mastery: dict[str, float], pool: list[dict]) -> dict:
    weakest = min(mastery, key=mastery.get)
    return next(c for c in pool if weakest in c["concepts"])
```

Alternative solution comparison

	Proposed	Alternative
approach	ml-based weakness tracking	rules-based tagging
question source	llm-generated plus curated	curated bank only
advantage	competitive advantage here	simpler, cheaper

Obstacles: external (PESTEL)

Political	
Economic	
Social	
Technological	
Environmental	
Legal	

Obstacles: internal

- technical: cold start problem, question quality
- human: motivation and retention
- resourcing: build time, llm costs
- mitigations for each

Business case: initiative and value

- type of initiative and why (startup, social enterprise or internal initiative)
- benefits and value delivered
- scale path: interview prep, ctf prep, corporate l&d

Business case: resources

- short term: people, tools, costs
- long term: people, tools, costs

Conclusion

- one line recap of problem, solution and value

References

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